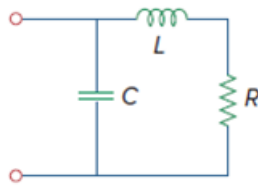


Problem

Find the resonant frequency of the circuit in Fig. 14.78.

Figure 14.78:



Step-by-step solution

Step 1 of 2

Refer to Figure 14.78 in the textbook.

In frequency domain inductor is represented as $j\omega L$, capacitor is represented as $\frac{1}{j\omega C}$.

In Figure 14.78, the inductor and resistor are in series and the combination is in parallel with the capacitor. Calculate the net admittance of the circuit.

$$\begin{aligned}
 Y &= j\omega C + \left(\frac{\frac{1}{j\omega L} \times \frac{1}{R}}{\frac{1}{j\omega L} + \frac{1}{R}} \right) \\
 &= j\omega C + \frac{1}{\frac{j\omega RL}{j\omega L + R}} \\
 &= j\omega C + \frac{1}{R + j\omega L} \times \frac{R - j\omega L}{R - j\omega L} \\
 &= j\omega C + \frac{R - j\omega L}{R^2 + \omega^2 L^2}
 \end{aligned}$$

Step 2 of 2

At resonance imaginary part of admittance is zero.

$$\operatorname{Im}(Y) = 0$$

$$\omega_0 C - \frac{\omega_0 L}{R^2 + \omega_0^2 L^2} = 0$$

$$C = \frac{L}{R^2 + \omega_0^2 L^2}$$

$$R^2 + \omega_0^2 L^2 = \frac{L}{C}$$

$$\omega_0^2 L^2 = \frac{L}{C} - R^2$$

$$\omega_0^2 = \frac{L}{CL^2} - \frac{R^2}{L^2}$$

$$\omega_0 = \sqrt{\frac{1}{CL} - \frac{R^2}{L^2}}$$

Hence, the resonant frequency of the circuit is

$$\sqrt{\frac{1}{CL} - \frac{R^2}{L^2}}.$$